Physics A (H156, H556)

Astrophysics 1
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	

Centre number						Candidate number				

INSTRUCTIONS TO CANDIDATES

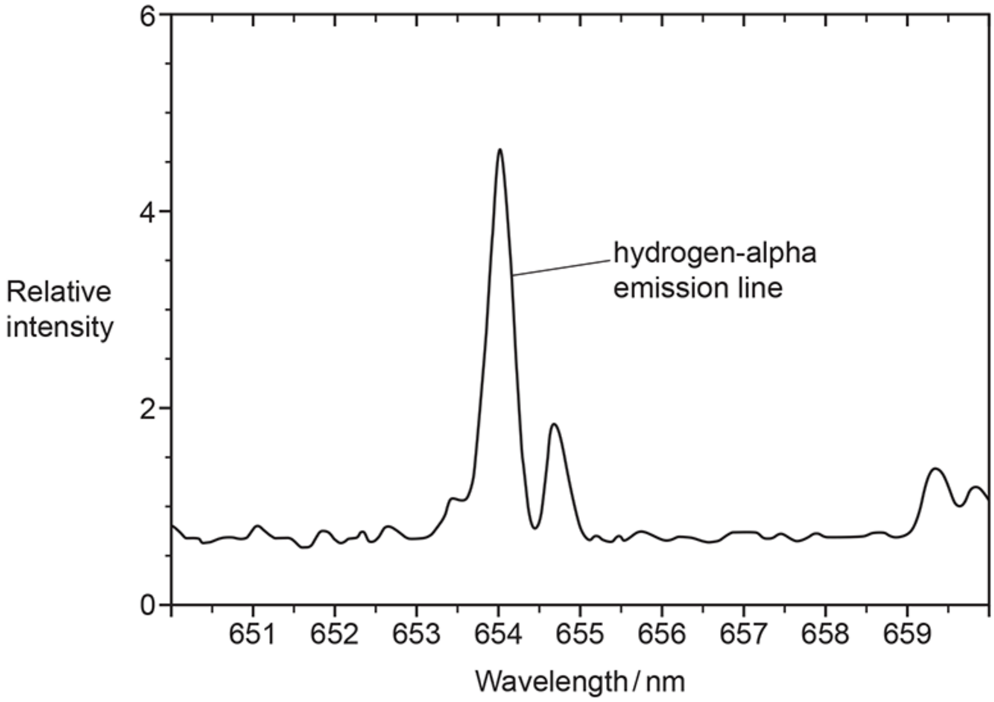
- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **50**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) The diagram below shows the part of the spectrum of light from the galaxy M87 with a strong hydrogen-alpha emission line.

The spectrum of a sample of hydrogen gas analysed in a laboratory on Earth showed this emission line to be at 656.3 nm.



Calculate the magnitude of the recession velocity of M87.

velocity = ... km s^{-1} **[3]**

(b) Estimate the distance in light years to M87 using Hubble's law.

Hubble constant $H_o = 2.5 \times 10^{-18}\ s^{-1}$

distance = ... ly **[2]**

(c) Estimate the percentage uncertainty in the distance calculated in part **(b)** using the width of the emission line in the diagram.

percentage uncertainty = +/– .... % **[2]**

(d) Other methods have shown M87 is actually 50 million light years from Earth.

Suggest why M87 might contradict Hubble's law.

[2]

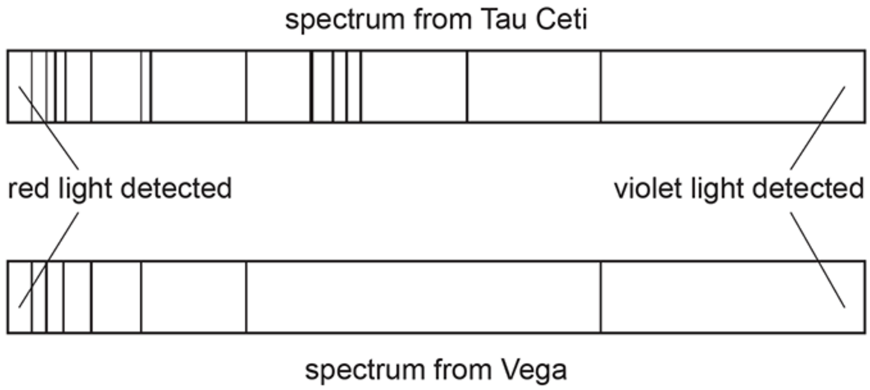
(e) Suggest **one** reason why the Big Bang theory is still the accepted model of the origin of the Universe.

[1]

2 Analysis of stellar light can identify the elements that are present in the outer layers of a star.

Absorption spectra of visible light collected from two main sequence stars, Vega and Tau Ceti, are shown below. The spectra were collected by a telescope orbiting the Earth. The dark lines on the diagram show wavelengths where the intensity was zero or close to zero.

- Explain the cause of the dark lines.
- Describe and explain at least one difference between the spectra.
- Analyse the spectra to determine which star is older.



[6]

3 Which of the following determines whether a super red giant evolves to a black hole?

A Dark matter
B Electron degeneracy
C Mass of the core
D The Chandrasekhar limit

Your answer [1]

4 Monochromatic light of wavelength 600 nm is passed through a diffraction grating.

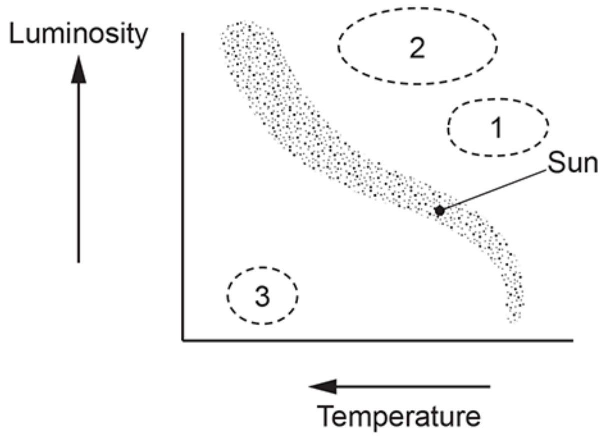
Maxima are formed on a screen 3.0 m from the grating. The distance between the two first order maxima, on either side of the zero maximum, is 1.42 m.

What is the spacing, in m, between lines on the diffraction grating?

A 1.3×10^{-6}
B 1.4×10^{-6}
C 2.5×10^{-6}
D 2.6×10^{-6}

Your answer [1]

5 Which option correctly gives the likely evolution of the Sun?

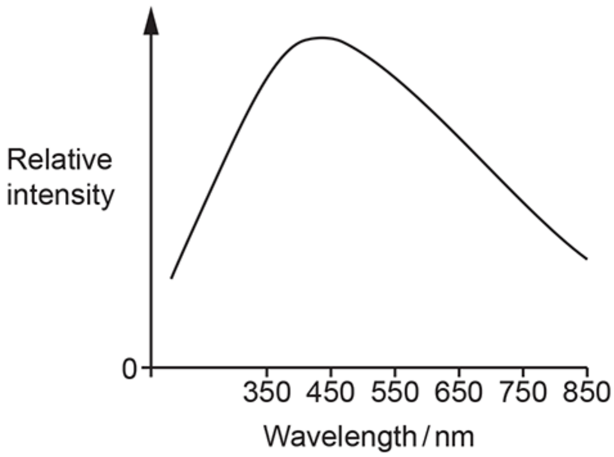


A $1 \rightarrow 2 \rightarrow 3$
B $1 \rightarrow 3$
C $2 \rightarrow 1 \rightarrow 3$
D $2 \rightarrow 3$

Your answer [1]

6 The surface temperature of the Sun is 5800 K and the peak wavelength is 500 nm.

What is the surface temperature, in K, of the star represented by the graph below?



A 3400
B 5200
C 6400
D 9900

Your answer [1]

7 Which of these statements about the Universe is implied by the Cosmological principle?

A Dark energy is found in every galaxy
B Kepler's laws can be applied to any solar system
C Matter and anti-matter were formed in the very early Universe
D Microwave background radiation corresponds to a temperature of 2.7 K

Your answer [1]

8 The apparent position of a star changes by an angle of 1.0×10^{-4} degrees between observations made 6 months apart from the same place on Earth.

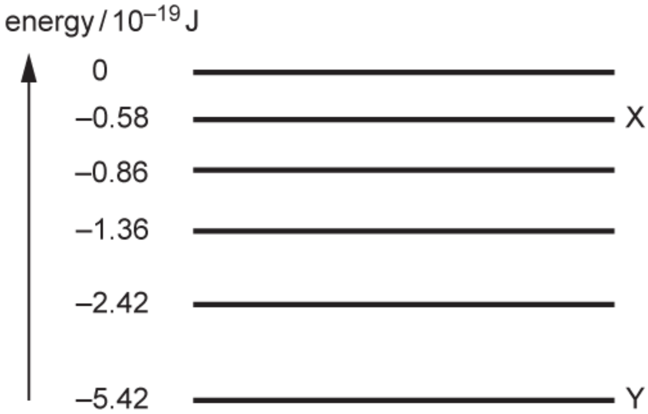
What is the distance of the star from Earth in parsecs?

A 2.8
B 5.6
C 170
D 330

Your answer [1]

9(a) The diagram shows some of the energy levels of the electron in a hydrogen atom.

energy / 10-19 J



An electron moves from energy level X to energy level Y.

Show that the wavelength of the photon produced is about 410 nm.

[2]

(b) The light from the stars in a distant galaxy is analysed on the Earth using a diffraction grating. Dark lines are observed in the spectrum.

An astronomer concludes that the dark line at a wavelength 432 nm corresponds to the electron transition between X and Y.

i. Explain the origin of the dark lines.

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[2]

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ii. Calculate the recession velocity *v* of the galaxy.

v = ... m s^{-1} **[2]**

iii. State the name of the theory that is supported by evidence from the measurement of the recession velocities of galaxies in the universe.

[1]

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10 Describe how to determine the wavelength λ of a monochromatic laser pointer using a diffraction grating.

As part of your answer, explain how to

- analyse the measurements collected using a graphical method.
- improve the accuracy of the measurements taken.

You are given the number of lines per mm for the diffraction grating.

[6]

11(a) This question is about analysing the electromagnetic radiation from the star Nu Persei in the Milky Way galaxy.

Fig. 21.1 shows the relative intensities of different wavelengths of electromagnetic radiation from Nu Persei.

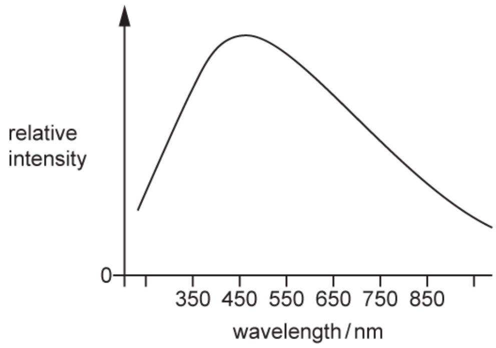

Fig. 21.1

The surface temperature of the Sun is 5800 K and its wavelength at which maximum intensity is emitted is 500 nm.

The luminosity of Nu Persei is 2.3×10^{29} W.

i. Use **Fig. 21.1** to show that the surface temperature of Nu Persei is about 6300 K.

[2]

ii. Estimate the radius of Nu Persei.

radius = ... m **[3]**

(b) This question is about analysing the electromagnetic radiation from the star Nu Persei in the Milky Way galaxy. Electromagnetic radiation is collected from Nu Persei by a sensor with an efficiency of 11% and cross-sectional area 1.0×10^{-4} m^2.

The radiant power collected by the sensor is 7.0×10^{-15} W.

i. Show that the radiant power per unit area arriving at the sensor is about 6×10^{-10} W m^{-2}.

[2]

ii. By the time the electromagnetic radiation from Nu Persei reaches Earth, the radiation from Nu Persei is evenly distributed over a spherical area with radius equal to the distance between Nu Persei and Earth.

Calculate the distance of Nu Persei from Earth in light years.

distance = ... light years **[4]**

(c) This question is about analysing the electromagnetic radiation from the star Nu Persei in the Milky Way galaxy. The luminosity of Nu Persei was estimated using the temperature of Nu Persei and the Hertzsprung-Russell (HR) diagram in **Fig. 21.2**. L is the luminosity of a star and L is the luminosity of the Sun.

The temperature data from earlier in this question is repeated in the table below.

Star	Surface temperature / K
Sun	5800
Nu Persei	6300

Comment on the uncertainty in your value, calculated in **part (ii) above**, of the distance of Nu Persei from Earth. You may write on the diagram as part of your answer.

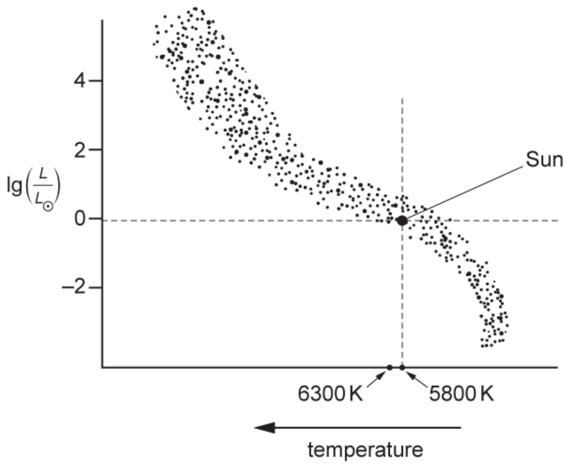

Fig. 21.2

--- **[3]**

12 Stars rotate around the centre of their galaxy.

Observations suggest that the stars at the edges of galaxies are moving at much higher velocities than expected.

What is the name given to the current explanation for these observations?

- **A** Chandrasekhar limit
- **B** Dark matter
- **C** The Cosmological principle
- **D** Wien's displacement law

Your answer **[1]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1	a		$\Delta\lambda$ = 656.3 - 654 (nm) or 2.3 (nm) $v = c \times \Delta\lambda / \lambda = 3 \times 10^8 \times 2.3 / 656.3$ $=1050$ km s^{-1}	C1 C1 A1	Allow between 2.25 and 2.35 Check denominator is 656.3 and not 654 (XP) ignore minus sign accept 1100 (km s^{-1}) **Examiner's Comments** The approach here is to find the wavelength corresponding to the peak of the hydrogen-alpha emission line and find the difference between that and the laboratory wavelength of the same line. The shift in wavelength as a fraction of the original wavelength is equal to the velocity of M87 as a fraction of the speed of light. Answers that used the wavelength 654 nm as the denominator were judged as 'XP' i.e. wrong physics, just as in previous series, even though this gave a very similar answer to the correct value for the velocity. Finally the unit on the answer line is km s^{-1}, so there was a conversion to complete. Most candidates completed all of these steps.
	b		$d=v/H_0 = 1050000 \div 2.5 \times 10^{-18}$= ($4.2 \times 10^{23}$ m) 4.2×10^{23} m $\div 9.5 \times 10^{15} = 4.4 \times 10^7$ light years	C1 A1	Ecf from (a) Using 1100 km s^{-1} gives 4.6×10^7 ly Accept use of H_o written in due to erratum Accept use of original H_o on paper i.e. 2.5×10^{-19} giving 4.4×10^8 (ly) or 4.6×10^8 (ly) if 1100 used. **Examiner's Comments** By using the candidate's value from part (a), quoted in ms^{-1} for this part, most candidates used an appropriate value for Hubble's constant to arrive at a distance in metres to M87. Once more there is a conversion to complete to get the value in the required unit. The mark scheme allows full credit for the correct use of the original value of H_0 if the erratum was not used, as well as use of any handwritten replacement value for H_0 used correctly.

Question			Answer/Indicative content	Marks	Guidance
	c		Absolute uncertainty in $\Delta\lambda$ in range of 0.4-1.2 nm % Uncertainty = (candidate's abs uncertainty ÷ 2.3) × 100= 21%	C1 A1	Allow value that if rounded to 1sf would be in the range Accept 1sf answer Allow in range 17 – 52 % **Examiner's Comments** This was a challenging item. Many candidates correctly found the width of the emission line according to the scale on the diagram. Others completed this incorrectly by using mm or cm. The percentage uncertainty in the distance is the same as the percentage uncertainty in the change in wavelength. Hence the correct expression to evaluate is the ratio of the width of the line (roughly 0.8 nm but there is a significant allowable range here) to the shift in the line (2.3 nm). Many candidates did not use the shift in the line but incorrectly used either the observed or the original wavelength instead.
	d		Two from: Hubble's law assumes/predicts all galaxies moving away / AW (from evidence in question) M87 is blue shifted/moving towards the Milky Way / AW idea that M87 may 'gravitationally associated' with Milky Way/Local cluster/Virgo supercluster Suitable comment comparing absolute uncertainty with difference in velocities or percentage uncertainty with percentage difference	B1 B1 B1 B1	i.e. correct statement of Hubble's law accept has a negative recession velocity allow any reasonable alternative e.g. percentage uncertainty from change in wavelength s larger than the percentage difference in the quoted distance and the calculated distance **Examiner's Comments** The mark scheme shows a wide variety of acceptable ideas that were creditworthy for this item. Many candidates provided one acceptable idea with some providing at least two.

Question			Answer/Indicative content	Marks	Guidance
	e		One from • Cosmic microwave background evidence • Very few galaxies are blue shifted	B1	Allow decipherable truncations or acyronyms e.g. CMBR, MBR Allow other reasonable ideas that supports model (e.g. nucleosynthesis ratios, He:H ratio enhancement (extra helium), universe may not be uniform and/or isotropic, no other model is consistent with the evidence, universe is still expanding) <u>Examiner's Comments</u> There were also a wide variety of acceptable candidate responses to this question.
			Total	10	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>2</td><td></td><td></td><td>Level 3 (5–6 marks)
Detailed explanation of dark lines. Differences between lines discussed with explanation. Correct conclusion with reasoning

There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.

Level 2 (3–4 marks)
some explanation of dark lines with some differences between lines

or

detailed description of dark lines

or

detailed description of the differences in line

There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.

Level 1 (1–2 marks)
EITHER
limited explanation of dark lines or a difference in lines given with an unsupported conclusion or no conclusion at all

There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.

0 mark
No response or no response worthy of credit.</td><td>B1 × 6</td><td>Indicative scientific points may include:
Dark lines

• Dark lines show wavelengths absorbed
• Star produces a continuous spectrum
• Atmosphere of star absorbs light
• Photons are absorbed by electrons increasing/moving ‘up’ energy levels
• Energy of photons match differences in energy levels of atoms
• Light is re-emitted by atoms in star atmosphere
• Light is re-emitted in random directions

Differences (number of lines and intensities)

• Tau Ceti has extra lines/Vega has fewer lines
• Lines in common are different intensities (stronger/weaker)
• Tau Ceti has more elements
• Different proportions of hydrogen.

Analysis/conclusion

• Vega is younger/Tau Ceti is older
• Tau Ceti has had more time to fuse heavier elements
• Tau Ceti has a lower proportion of hydrogen because it has been fusing/using it for longer.
• Accept that idea Vega may be older than Tau Ceti if Tau Ceti has more mass

<u>Examiner’s Comments</u>

Many candidates spotted that the spectral lines from Vega exactly matched up to the corresponding lines in the spectrum from Tau Ceti. This rules out the idea that one star’s spectrum has been red shifted by more than the other, as suggested by many candidates. The principal difference is that Tau Ceti has more lines because its atmosphere has more elements in it than that of Vega. A small number of candidates also spotted that one of the lines in Tau Ceti’s spectrum is darker than the rest.

Many candidates produced exceptionally detailed explanations of how absorption</td></tr>
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Question			Answer/Indicative content	Marks	Guidance
					spectra occur. Very few attributed the spectra to such phenomena as destructive interference or something to do with the Earth's atmosphere. The key to accessing Level 3 for this question lay in explaining why the two spectra were different. Many candidates suggested that there were more elements in the atmosphere of Tau Ceti because it had been fusing hydrogen for so long that some of those products were also fusing together to produce even heavier elements. A few candidates linked the darker line in Tau Ceti's spectrum to a large abundance of an element that wasn't hydrogen. Exemplar 3 An absorption Spectrum is a series of dark lines on a continuum Spectrum due to light being passed through a cold low density gas such as the Photosphere of a Star, causing the electrons in the atoms to absorb the photons and be excited. The position of the dark lines correspond to the frequency of the photons absorbed. Each element has a unique Spectrum due to the unique energy levels in their atoms, as to excite ~~a photon~~ an electron, the energy of the photon must be equal to the difference between energy levels. The Spectrum of Tau ceti has more lines than Vega, and all of the lines in Vega's Spectrum align with Tau ceti. This means that Tau ceti's Photosphere contain elements that are not present in Vega's. Additionally Vega's spectrum seems to be redshifted as the lines ~~one~~ are more to the left and more spaced out. This could be because, according to Hubbles law, Vega is futher away and is receding with a greater Speed, $V=H_0D$. CR 2025 Tau ceti is older because the Spectra Shows that there are Many More elements in the star, indicating that Tau ceti is no longer just fusing hydrogen into helium, but is fusing ~~a~~ heavier elements. Tau ceti May be a red giant and Vega May be a Main sequence star. The first paragraph of this response is a detailed description of how continuous light from a star can be selectively absorbed by the atmosphere of that star. The candidate doesn't add in the details about de-excitation or of re-emission of the light in random directions; however, they have included everything up to that point. The candidate has also spotted that there are more lines for Tau Ceti than for Vega and attributes this to elements that are present in the photosphere of Tau Ceti that are not present in that of Vega. The argument about red shift is incorrect. Having said that, the candidate has gone on to mention about the fusion products other than helium and attributes this to Tau Ceti being the older star. At the end,

Question			Answer/Indicative content	Marks	Guidance
					the mention of Tau Ceti *maybe* being a red giant and no longer main sequence is incorrect (the question says that both stars are main sequence). On balance, this response was awarded Level 3.
			Total	6	
3			C	1	<u>Examiner's Comments</u> Many students remembered that the mass of the core is the crucial factor in determining whether a super red giant evolves into a black hole.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
4			D	1	<u>Examiner's Comments</u> Candidates often confused which formula to use. Some used the two-slit formula and others, correctly, used the diffraction grating formula. Here using *dsinq* = *nl* is the correct approach. The question gives the distance between the two first-order maxima. This means that the distance from the centre to the first-order maximum is *half* of that i.e. 0.71 m. Consequently, the angle q is $\tan^{-1}(0.71 \div 3.0) = 13.3....$ degrees. Substituting in the wavelength (600×10^{-9} m) and n=1 gives d as 2.61×10^{-6} m. Therefore the answer is D.
			Total	1	
5			B	1	<u>Examiner's Comments</u> Most candidates remembered that the correct sequence for the Sun is main sequence, red giant, white dwarf (i.e. answer B). Any answer that includes red super giant cannot be correct.
			Total	1	
6			C	1	<u>Examiner's Comments</u> This question tests knowledge of Wien's Law, $l_{max}T$ = constant. Using the given data for the Sun, the constant here is 2.9×10^6 nm K. The wavelength of maximum intensity on the graph is roughly 450 nm so the temperature = $2.9 \times 10^6 \div 450$ = 6444 K, giving the correct answer as C.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
7			B	1	<u>Examiner's Comments</u> The Cosmological principle is that on a large scale, the universe is homogeneous (the same everywhere) and isotropic (the same in all directions). Answers A, C and D are not implied by this, although they may be true statements. Answer B is correct here.
			Total	1	
8			B	1	<u>Examiner's Comments</u> The data in the question gives a parallax angle of 0.5×10^{-4} degrees. This equates to 0.18 arcseconds. Parallax angle and distance in parsecs are inversely proportional, with 1 arcsecond giving a distance of 1 parsec. 0.18 arcseconds gives a distance of (1 ÷ 0.18) = 5.6 parsecs.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
9	a		Difference in energy level = 4.84 (× 1^{-19}) (J) $\lambda = hc/4.84 \times 10^{-19}$ and 4.11×10^{-7} (m) seen	M1 A1	Accept (–.58 – (–5.42)) for 4.84 Accept E = hf to find f, followed by $\lambda = c/f = 4.11 \times 10^{-7}$ (m) seen Allow 4.104×10^{-7} (m) seen <u>**Examiner's Comments**</u> As with other 'show that' questions, many candidates understood what to do to find their value of the wavelength, without necessarily showing sufficient evidence of formula re-arrangement.
	b	i	Dark lines are **absorption** lines / idea of photon absorption / idea of electrons absorbing energy / idea of electrons making transition to higher energy level (i.e. excitation) Caused by light passing through (cooler) stellar atmosphere or Earth's atmosphere	B1 B1	Allow (cooler) gases/hydrogen (gas) <u>**Examiner's Comments**</u> Apart from a small number of candidates that confused emission spectra with absorption spectra, it was clear that most candidates recognised that what they needed to do. Technical language skills prevented some otherwise sensible answers from scoring 2 marks.
		ii	$\Delta\lambda/\lambda$ = 22/410 or 21/411 v= 0.054c = $1.6(1) \times 10^{7}$ (m s^{-1})	C1 A1	NB 21/411 gives 1.5×10^{7} (m s^{-1}) Any evidence of use of 432 as denominator of $\Delta\lambda/\lambda$ is XP zero marks <u>**Examiner's Comments**</u> Most candidates selected the correct formula in this part. If a candidate responded at all, they were likely to score both marks. A small percentage of candidates used the wrong wavelength for the denominator of the relevant expression. This was marked as 'wrong Physics' (XP) on scripts.
		iii	Big Bang / Expanding universe i.e. Hubble's Law / dark matter or dark energy / AW	B1	ignore redshift ignore Doppler <u>**Examiner's Comments**</u> An overwhelming majority of candidates made some reference to either the Big Bang Theory, cosmological expansion or simply 'Hubble's Law'. Some other answers were also given credit as shown in the Mark Scheme although these were very much rarer.

Question			Answer/Indicative content	Marks	Guidance
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
10			**Level 3 (5–6 marks)** description and analysis, with an appropriate comment about accuracy *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Description and some graphical analysis and some comment about accuracy *There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Incomplete description or analysis, or an appropriate comment about accuracy *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 mark** *No response or no response worthy of credit.*	B1 × 6	NB - use of two slit interference formula is incorrect physics. Mark XP and treat description as incomplete **Indicative scientific points may include:** **Description** • Angles or distances measured from central maximum • Some method of measuring angle directly, e.g. protractor • Measuring distances from grating to screen and from central maximum to others • Description of maxima • White screen or wall • Recognition and mitigation of risk of injury to eye **Analysis** • Graph of d sin theta against n with gradient lambda (or workable alternative) • Conversion of lines per mm to distance between lines **Accuracy** • Use large room to achieve large distances • Large distance reduces relative uncertainty • Measure distance between corresponding maxima and halve • Use of trigonometry to convert measured distances into angles from the centre • use spots either side of the centre • change diffraction grating for smaller d so that maxima are more spaced out • change diffraction grating for larger d so that there are more maxima (and so more points on the graph) • darken the room <u>**Examiner's Comments**</u> This experiment is one of the suggested practicals to complete in the Practical Endorsement. The relevant formula was not on the question paper itself, however it is in the relationships and formulae booklet in the appropriate place. Candidates that described the two-slit

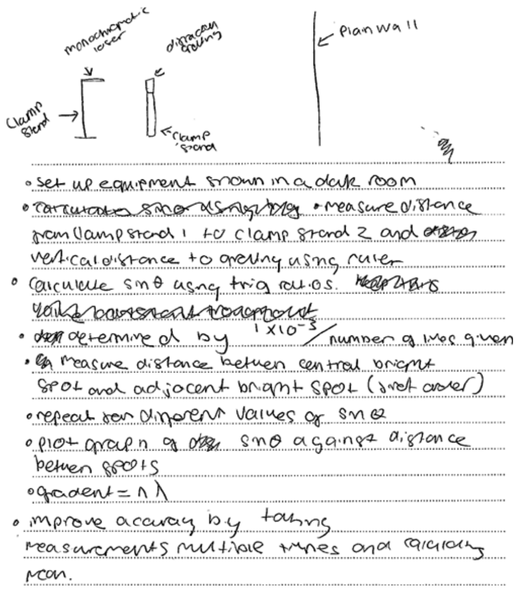
Question			Answer/Indicative content	Marks	Guidance

interference experiment could not, therefore, be awarded any more than Level 1 (2 marks).

As explained in previous series, approaches required a graphical element to find the wavelength (amongst other aspects) in order to achieve Level 2 (4 marks).

Finally, a key aspect that distinguished between candidates was their ability to describe and explain the necessary precautions for high quality data. A selection of these is listed in the Mark Scheme.

Exemplar 1



Exemplar 1 has satisfied all of the criteria for a Level 3 response, 6 marks.

The candidate has described the correct experiment and made it clear how they are going to measure the relevant quantities such as the distance of the maxima away from the centre and the distance to the screen so that they can later calculate theta and hence sin(theta). This already is promising for a Level 3 response.

The candidate has also completed the analysis correctly by using a graphical approach. It is clear what is going to be plotted on which axis and what the gradient of the line of best fit will represent. Finally, it is clear how the

Question			Answer/Indicative content	Marks	Guidance
					wavelength shall be calculated. The candidate has used trigonometry to calculate theta and has stated that the room should be dark, All of these factors taken together indicate Level 3 thinking.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
11	a	i	Estimate of λ_{max} between 440 and 470 nm (T =) 5800 x 500/460 = 6300 K i.e. candidate's value	M1 A1	Allow 2.9 million for 5800 x 500 Allow assumption of 6300 K, calculation of lambda (M1) and then confirmation on graph (A1) for 2 marks **Examiner's Comments** As in previous 'show that' questions, most candidates started this question well by obtaining λmax from the graph and calculating the surface temperature. This was insufficient as a consistent approach to showing working for these questions was applied
		ii	$L = 4\pi r^2 \sigma T^4$ Correct substitution of values i.e. $2.3 \times 10^{29} = 4\ \pi\ r^2\ \sigma\ 6300^4$ radius = $1.4(3) \times 10^{10}$ (m)	C1 C1 A1	**Examiner's Comments** As long as candidates used the equation for Steffan's Law, they tended to score well on this item. There were some candidates that transcribed the power of 4 incorrect yet had ECF applied to their work to score most of the marks available.
	b	i	$P = 7.0 \times 10^{-15} \div 0.11$ $I = 6.36 \times 10^{-14} / 1.0 \times 10^{-4} = 6.4 \times 10^{-10}$ ($\mathrm{Wm^{-2}}$)	M1 A1	**Examiner's Comments** Most candidates successfully manipulated the data given to show that the radiant power per unit as required.

Question			Answer/Indicative content	Marks	Guidance
		ii	$L/4\pi r^2 = 6.4 \times 10^{-10}$ $r = (L \div (6.4 \times 10^{-10} \times 4\pi))^{0.5}$ $=5.4 \times 10^{18}$ (m) ($\div 9.5\times10^{15}$ =) 560 light years	C1 C1 C1 A1	Allow use of 6×10^{-10} Allow alternative method: Finding intensity at star surface C1 Using ratio of intensities = square of ratio of (star radius / distance to earth) C1 Use of 6×10^{-10} for I gives 580 ly <u>Examiner's Comments</u> This is a challenging multi-step question. Essentially it tests understanding of the inverse-square relationship. Some candidates simply used the formula given on the formula sheet. Others did not, however successfully manipulated the data to use the inverse-square rule and/or finding the intensity of radiation leaving the surface of the star. Once the candidate arrived at the distance in metres, it was relatively simple to find the distance in light years.

Question			Answer/Indicative content	Marks	Guidance
	c		Any three from: There is a range/ 'band' in the luminosity values at 6300 K range of luminosity is between L. and about 10 L. The uncertainty in the distance calculated = ½ the uncertainty in the luminosity AW Reference to Nu Persei not necessarily being on the main sequence	B1 B1 B1 B1	Check diagram for rewardable content <u>Examiner's Comments</u> This item also proved very challenging. The key idea here is that there is a range of luminosities (the inherent power of the star) that correspond to a temperature of 6300 K, not least that the star may not be main sequence at all. These ideas are reminiscent of a question in a previous paper. After those initial ideas, the candidates needed to discuss this more quantitatively by noticing that the range of luminosities was about 10, given that the scale is logarithmic. **Assessment for learning** There is a lot of Physics relevant to the H-R diagram other than merely the shape of the main sequence and the positions of other classifications of stars. Teachers might consider explaining how the data for the H-R diagram is collected and processed, along with the accompanying uncertainties in those measurements.
			Total	14	
12			B	1	<u>Examiner's Comments</u> The Chandrasekhar limit relates to white dwarf stars and their evolution. Wien's displacement law relates the surface temperature of a star and λ_{max}. The Cosmological principle has nothing to do with stars' velocity in a galaxy, hence the correct answer is B.
			Total	1	

